

FADA N'GOURMA, Burkina Faso

WMO: 655070

Lat: 12.033N

Lon: 0.367E

Elev: 309

StdP: 97.67

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.2	18.3	-7.1	2.1	29.5	-5.2	2.5	28.5	6.4	25.8	5.9	27.6	1.4	40	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.2	40.6	21.0	39.7	21.1	38.9	21.1	27.1	32.4	26.7	32.2	26.3	31.9	2.4	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.7	21.8	29.8	25.2	21.2	29.4	25.0	20.8	29.0	87.2	32.4	85.4	32.3	83.9	32.0	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.5	DB	14.8	42.5	1.3	0.3	13.9	42.7	13.1	42.9	12.4	43.1	11.5	43.4	(4)
(5)			WB	7.0	27.8	0.7	0.5	6.5	28.2	6.1	28.5	5.7	28.8	5.2	29.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	25.7	28.8	32.5	33.4	31.8	29.5	27.2	26.4	27.2	29.0	28.8	26.2	(6)
(7)		DBStd	3.03	2.12	2.37	1.55	1.87	2.20	2.04	1.68	1.30	1.56	1.23	1.31	1.79	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	6891	487	527	698	702	677	586	534	508	517	588	563	503	(10)
(11)		CDD18.3	3849	229	294	439	452	419	336	276	250	267	329	313	245	(11)
(12)		CDH23.3	48901	2499	3823	6822	7117	6207	4353	2777	2173	2623	3864	3949	2693	(12)
(13)	CDH26.7	27070	1250	2221	4472	4763	3826	2239	1025	647	1003	1924	2319	1382	(13)	
(14)	Wind	WSAvg	1.9	2.0	2.0	1.8	2.3	2.6	2.3	1.9	1.7	1.3	1.4	1.3	1.9	(14)
(15)	Precipitation	PrecAvg	823	0	1	6	28	68	115	183	232	150	36	1	1	(15)
(16)		PrecMax	1080	2	12	40	139	140	215	388	448	332	153	13	39	(16)
(17)		PrecMin	554	0	0	0	0	10	27	90	113	1	0	0	0	(17)
(18)		PrecStd	134	0	2	10	31	34	39	65	80	63	31	3	6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	40.0	41.3	41.7	40.5	37.7	35.2	32.7	34.6	37.2	38.1	36.9	(19)
MCWB			17.2	19.2	19.5	21.9	24.2	25.4	25.3	26.4	26.4	21.9	19.7	18.2	(20)	
2%		DB	35.6	38.6	40.5	40.7	39.2	36.4	33.5	31.7	33.4	36.2	37.4	35.6	(21)	
		MCWB	16.6	18.7	19.8	21.4	24.7	25.4	25.5	26.0	26.4	22.9	19.7	17.7	(22)	
5%		DB	34.2	37.4	39.7	39.9	38.0	35.3	32.3	30.8	32.4	35.1	36.7	34.4	(23)	
		MCWB	16.1	18.3	19.8	21.7	24.8	25.3	25.5	25.7	26.4	24.1	19.7	17.1	(24)	
10%		DB	32.8	36.0	38.9	39.0	36.9	34.2	31.1	29.9	31.4	34.0	35.8	33.1	(25)	
		MCWB	15.6	17.7	19.6	22.1	25.0	25.2	25.3	25.5	26.2	24.7	19.6	16.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	23.9	25.0	26.1	26.8	26.9	26.7	27.2	27.6	27.4	24.9	22.6	(27)
MCD B			32.1	34.5	34.8	36.2	34.7	33.7	32.0	31.7	32.2	33.1	33.0	29.6	(28)	
2%		WB	17.8	22.3	24.2	25.6	26.3	26.4	26.2	26.5	27.1	26.8	24.1	19.4	(29)	
		MCD B	33.1	31.2	35.0	35.5	34.7	33.1	31.2	30.6	31.8	32.5	32.3	32.1	(30)	
5%		WB	16.9	20.4	23.6	25.1	26.0	26.0	25.8	26.1	26.7	26.4	23.2	18.3	(31)	
		MCD B	32.7	34.1	34.9	34.9	34.7	32.8	30.6	29.9	31.2	32.0	31.2	32.3	(32)	
10%		WB	16.0	18.6	22.7	24.7	25.7	25.7	25.5	25.7	26.3	26.0	22.3	17.2	(33)	
		MCD B	31.9	34.7	34.5	34.3	34.1	32.1	30.0	29.1	30.6	31.5	30.8	31.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	14.2	14.2	13.3	11.2	9.8	8.7	7.3	6.8	8.0	10.8	15.4	14.7	(35)	
5% DB		MCD BR	16.0	15.6	14.3	12.6	11.2	9.9	9.0	8.2	9.5	12.4	15.9	15.7	(36)	
		MCWBR	6.7	6.3	5.4	4.3	2.7	2.5	2.9	3.3	3.8	3.7	5.0	6.1	(37)	
		MCD BR	15.8	14.0	11.9	10.6	10.0	9.0	8.0	7.8	9.0	10.2	13.3	15.4	(38)	
(39)		5% WB	MCWBR	7.1	6.4	5.4	3.4	2.9	3.0	3.1	3.6	4.0	4.0	4.7	6.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.604	0.669	0.763	0.736	0.682	0.636	0.572	0.547	0.547	0.618	0.562	0.546	(40)	
taud		1.716	1.581	1.444	1.578	1.647	1.752	1.948	2.006	1.988	1.796	1.812	1.831	(41)		
Ebn at Noon		710	684	633	649	672	694	744	774	779	713	739	746	(42)		
Edh at Noon		233	275	321	279	256	227	188	180	184	219	209	203	(43)		
(44)	All-Sky Solar Radiation	RadAvg	5.54	5.84	6.05	6.21	6.02	5.68	5.33	5.11	5.50	5.88	5.68	5.45	(44)	
(45)		RadStd	0.28	0.28	0.28	0.20	0.21	0.20	0.15	0.18	0.25	0.18	0.14	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page